

Lactic Acid Fermentation

Lactic acid is produced by numerous types of bacteria that use anaerobic respiration to break down sugars into energy in the absence of oxygen. Lactic acid fermenters typically are active in aqueous (wet) environments in which oxygen is limited or absent. Lactic acid-fermenting bacteria, as well as other types of bacteria and yeasts that perform anaerobic respiration, are responsible for the flavor, texture, and nutrition of fermented foods. More important for preindustrial societies, however, was the benefit to the shelf life of fermented foods, which can extend the period in which a food is edible from hours to days, weeks, and even years (in the case of some cheeses).

The examples below illustrate the diversity of foods that are produced by lactic acid fermentation (sometimes in concert with other fermentation pathways) and have been used for thousands of years to extend the shelf lives of foods as varied as flour, milk, fruits, and vegetables.

Bread

Today, most bakeries make bread from only four ingredients: flour (usually of wheat), water, yeast, and salt. The use of this single yeast, known as baker's yeast (a strain of *Saccharomyces cerevisiae*), is a relatively recent phenomenon of the industrial era. Traditional methods of producing leavened breads (that is, those that rise) use what we today call sourdough methods: using a starter that is a culture of multiple yeasts and bacteria to initiate the fermentation that produces the breads we recognize today. Sourdough starters can be quite diverse, with different starters mixing different bacterial strains—primarily lactic acid fermenters—and species of yeast. It is possible to maintain a sourdough starter indefinitely if it is fed regularly with flour and water, and it is possible to create a new starter from only those ingredients, using bacteria and yeasts that naturally occur on the surface of the grain before grinding.

Sourdough breads benefit from both lactic acid-fermenting bacteria and yeasts breaking down complex carbohydrates (i.e., starches) in flour and producing both simple and complex byproducts. Carbon dioxide is produced in abundance by both bacteria and yeasts during fermentation, giving breads the leavening that allows them to rise and expand. However, that carbon dioxide can either escape the bread, giving rise to a dense bread (such as a German or Danish rye bread), or be trapped by networks of protein (especially gluten, which is found in high quantities in bread wheat, which we use for most bread flours today), producing an open crumb and a large, airy loaf.





A dense, seeded rye bread typical of northern Europe; this is an example of a German Vollkornbrot (literally, “whole grain bread”), with whole rye seeds (as well as millet and pumpkin seeds) added to a 100% whole rye flour sourdough bread. Note the few, very small bubbles; rye contains a particularly weak set of proteins (it is considered “low-gluten”).

Source: Dr. Marston

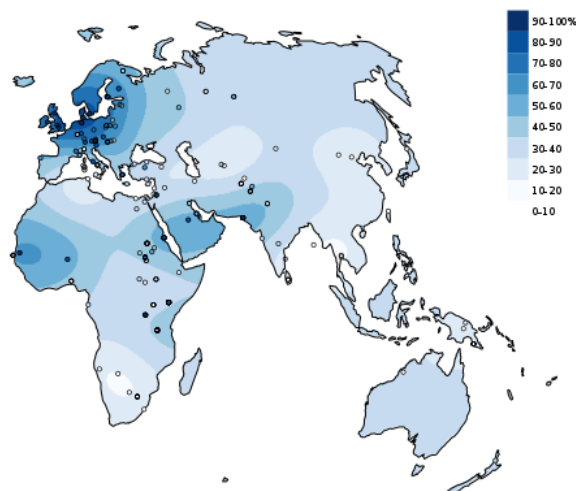


A sourdough wheat bread, made from a mixture of high-gluten bread wheat (*Triticum aestivum*) flour and moderate-gluten emmer wheat (*Triticum dicoccum*) flour. Note the network of fine holes that are the result of carbon dioxide bubbles trapped in the protein network of the loaf.

Source: Dr. Marston

Dairy

Most humans are lactose intolerant following weaning (cessation of breast feeding); as mammals, we evolved to make use of milk only during our infancy. Some populations, however, have evolved lactase persistence, the ability to make lactase (the enzyme that breaks down the sugar lactose) into adulthood. As indicated in the map below, lactase persistence is only common in Europe and areas of the African Sahel, Arabia, and the Indus Valley. Notably, these are areas in which cattle were domesticated and/or where dairying formed a major component of historical subsistence strategies.



Frequency of lactase persistence (indicated by darker blue colors) among ancestral populations of Eurasia, Africa, Australia, and Oceania.

Source: [Wikimedia Commons](#)

(https://commons.wikimedia.org/wiki/File:Lactose_tolerance_in_the_Old_World.svg)

One solution to widespread lactose intolerance that allows the consumption of dairy products is the use of fermentation to convert milk into fermented dairy products: yogurts and cheeses. By deliberately introducing specific bacteria to milk, it is possible to predictably produce fermented milk products. One easy way to do this is to add small amounts of a starter to milk, which is often first heated to denature its proteins and allow bacteria more ready access to the nutrients they need for growth. For example, yogurt can be easily made at home: First, heat milk to 180–185° F (85° C), then cool it to 115° F (45° C) and mix in a tablespoonful of yogurt. The milk must then incubate at warm temperatures (around 105–115° F, 40–45° C) for 6–24 hours to culture the bacteria. After this time, the milk is semisolid and sour (this is a lactic acid fermentation process, primarily driven by the bacteria *Lactobacillus delbrueckii* and *Streptococcus thermophilus*) and can be further fermented into cheese, strained to make “Greek”-style yogurt, or watered down and salted to make ayran (<https://en.wikipedia.org/wiki/Ayran>) or similar beverages.

Cheeses offer even more longevity than yogurts, as they can be hard and encased in a thick, dense rind that excludes pathogens. Because hard cheeses have lost most of their water content, they are stable at room temperature and can be stored and transported over long distances. Picture the giant wheels of Parmesan cheese in Whole Foods; this cheese can be stored in large blocks for years. Cheeses offer additional nutritional benefits: They concentrate fat and protein from milk, so

they are much more calorie dense than watery milk, and the bacteria, yeasts, and molds used in cheese production help to break down lactose and other indigestible molecules, rendering cheeses more palatable to those who are lactose intolerant.



Parmigiano Reggiano cheeses aging; these cheeses were three years old when photographed.

Source: [Wikimedia Commons](https://commons.wikimedia.org/wiki/File:Parmigiano_reggiano_factory.jpg)

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Chocolate

Make sure you have read the very short section of the textbook by Joan Slonczewski and John W. Foster on chocolate fermentation. Realize that even though chocolate has been produced for thousands of years and is one of the most economically important foods produced today, scientists still do not fully understand the complex suite of fermentation processes needed to convert ripe cacao pods into seeds suitable for roasting and grinding to produce chocolate!

Observe that Figure 2B in Slonczewski and Foster indicates the relative importance of multiple types of fermentation processes that occur over the roughly seven days it takes to ferment cacao. Yeasts and lactic acid bacteria dominate the first two days of fermentation, acetic acid fermenters take over by day three but entirely die out on day five, and spore-forming bacteria grow slowly throughout the process to take over fermentation in the final days of cacao processing. This type of multistage fermentation is common in complex fermentation processes, including some types of cheese production.

Pickles

Natural pickling is a lactic acid fermentation process in which bacteria that naturally occur on vegetables multiply and produce an increasingly low pH, resulting in a strongly acidic solution that inhibits the growth of competing bacteria, molds, and yeasts. The primary bacterium responsible

for pickling, *Lactobacillus plantarum*, can survive in a low-pH environment and continue to produce lactic acid, driving the pH even lower. Pickling is an effective mechanism for preserving nearly any fruit or vegetable, although some are more appealing when pickled than others.

One of the easiest foods to ferment is cabbage. Sauerkraut and kimchi are only two of the most famous types of fermented cabbage, and both are produced the same way. You can try to make your own sauerkraut at home! It's easy and only requires a few tools, such as a clean glass jar, and